# BUSFIN 8200: Problem Set 2

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## Question 1

### (a)

We have complete markets iff we can find a unique matrix $\Omega$ s.t. $X\Omega = I$, where

$$X = \begin{bmatrix} 10 & 10 & 0 \\ 20 & 10 & 10 \\ 30 & 10 & 10 \end{bmatrix}, \qquad \Omega = \begin{bmatrix} | & | & | \\ w_1 & w_2 & w_3 \\ | & | & | \end{bmatrix},$$

and each $w_i$ is a portfolio weight vector using A, B and C. For example, $(10, 20, 30)'$ is asset A's payoff across the three states.

$X\Omega = I$ means

$$Xw_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad Xw_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad Xw_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Because $X$ is $3 \times 3$, $X\Omega = I$ has a unique solution iff $X$ is invertible, and

$$X \text{ invertible} \iff \det(X) \neq 0 \iff \operatorname{rank}(X) = 3.$$

$$X = \begin{bmatrix} 10 & 10 & 0 \\ 20 & 10 & 10 \\ 30 & 10 & 10 \end{bmatrix} = 10 \begin{bmatrix} 1 & 1 & 0 \\ 2 & 1 & 1 \\ 3 & 1 & 1 \end{bmatrix}.$$

Row reduction:

$$\begin{aligned} \text{row } 3 - \text{row } 2: &\quad \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 3 & 1 & 1 \end{bmatrix} \\ \text{exchange rows 1 and 2}: &\quad \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 3 & 1 & 1 \end{bmatrix} \\ \text{row } 2 - \text{row } 1: &\quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 1 & 1 \end{bmatrix} \\ \text{row } 3 - 3 \cdot \text{row } 1 - \text{row } 2: &\quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

Thus $\operatorname{rank}(X) = 3$ and $X$ is invertible. $\Omega = X^{-1}$ exists and is unique, with $X\Omega = XX^{-1} = I$. Thus, the market is complete.

## (b)

Let $X$ be the matrix of $N = 3$ non-redundant asset payoffs, let $P$ be the price vector of the original assets, and let $P_c = (q_1, q_2, q_3)'$ be the price vector of the contingent claims. Use the LOP to solve for $P_c$ as a function of $X$ and $P$:

$$P = \begin{pmatrix} 16 \\ 8 \\ 6 \end{pmatrix}, \qquad X = \begin{bmatrix} 10 & 10 & 0 \\ 20 & 10 & 10 \\ 30 & 10 & 10 \end{bmatrix}.$$

The LOP says that two assets with identical payoffs in every state of nature must have the same price. Therefore,

$$X'P_c = P \quad \Longrightarrow \quad \begin{bmatrix} 10 & 20 & 30 \\ 10 & 10 & 10 \\ 0 & 10 & 10 \end{bmatrix} \begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix} = \begin{pmatrix} 16 \\ 8 \\ 6 \end{pmatrix}.$$

Dividing everything by 10:

$$\begin{cases} q_1 + 2q_2 + 3q_3 = 1.6 \\ q_1 + q_2 + q_3 = 0.8 \\ q_2 + q_3 = 0.6 \end{cases} \quad \Longrightarrow \quad \begin{cases} q_1 = 0.2 \\ q_2 = 0.4 \\ q_3 = 0.2 \end{cases}$$

Thus, $P_c = (0.2, 0.4, 0.2)'$.

## (c)

$$P_D = \text{Payoff}(D) \cdot P_c = \begin{pmatrix} 30 & 20 & 10 \end{pmatrix} \begin{pmatrix} 0.2 \\ 0.4 \\ 0.2 \end{pmatrix} = 6 + 8 + 2 = 16.$$

## (d)

$$P(X) = E[MX] = \sum_{s=1}^{S} \pi(s) M(s) X(s).$$

Since $q_s = \pi(s)M(s)$, we have $M(s) = q_s/\pi(s)$. With

$$P_c = \begin{pmatrix} 0.2 \\ 0.4 \\ 0.2 \end{pmatrix}, \qquad \pi(s) = \begin{pmatrix} 1/4 \\ 1/2 \\ 1/4 \end{pmatrix} \quad \Longrightarrow \quad M(s) = \begin{pmatrix} 0.8 \\ 0.8 \\ 0.8 \end{pmatrix}.$$

By definition, no arbitrage means that

1. An asset with payoff $\geq 0$ must have price $\geq 0$.
2. An asset with payoff $\geq 0$ and a chance for payoff $> 0$ must have price $> 0$.

We have $X \geq 0$, $\Pr(X > 0) > 0$, and $P(X) = E[MX] > 0$ (since $M > 0$). Thus, there is no arbitrage.

Moreover, $P(X) = E[MX] = 0.8\,E[X]$. The SDF stays constant across the three states, which implies it is risk-neutral.

**(e)**

$$\begin{aligned} P_D = E[MX_D] &= \sum_{s=1}^{S} \pi(s) M(s) X_D(s) \\ &= \tfrac{1}{4} \cdot 0.8 \cdot 30 + \tfrac{1}{2} \cdot 0.8 \cdot 20 + \tfrac{1}{4} \cdot 0.8 \cdot 10 \\ &= 6 + 8 + 2 = 16. \end{aligned}$$

This is the same as in Q1(c).

# Question 2

## (a)

I construct the unconditional linear SDF as

$$\hat{M}_t^* = \hat{a} - \hat{b}' x R_t,$$

where $\hat{b} = \hat{\Sigma}^{-1}\hat{\mu}$ and $\hat{a} = 1 + \hat{b}'\hat{\mu}$, imposing the normalization $E[M^*] = 1$.

Table 1 reports the estimated SDF coefficients. The estimated SDF satisfies the unconditional pricing conditions, with $\widehat{E}[\hat{M}^*] = 1$ and $\widehat{E}[\hat{M}^* x R_j] \approx 0$ for all 10 portfolios.

Figure 1 shows no clear monotonic relationship between the estimated $b_j$ coefficients and book-to-market deciles. Instead, the coefficients fluctuate considerably across portfolios, with both positive and negative values.

This pattern arises because the coefficients are determined by both the expected excess returns and the covariance structure of the portfolios, as reflected in $\hat{b} = \hat{\Sigma}^{-1}\hat{\mu}$. Therefore, portfolios with higher average excess returns do not necessarily receive larger positive coefficients.

Table 1: Estimated BM SDF intercept and coefficients, rounded to four decimal places. Each $\hat{b}_j$ corresponds to BM decile $j$; estimation uses decimal monthly excess returns.

| $\hat{a}$ | $\hat{b}_1$ | $\hat{b}_2$ | $\hat{b}_3$ | $\hat{b}_4$ | $\hat{b}_5$ | $\hat{b}_6$ | $\hat{b}_7$ | $\hat{b}_8$ | $\hat{b}_9$ | $\hat{b}_{10}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 1.0368 | -1.5850 | 3.5503 | 0.7104 | -3.1122 | 1.1837 | 2.9972 | -5.2931 | 2.4722 | 2.8775 | -1.1771 |

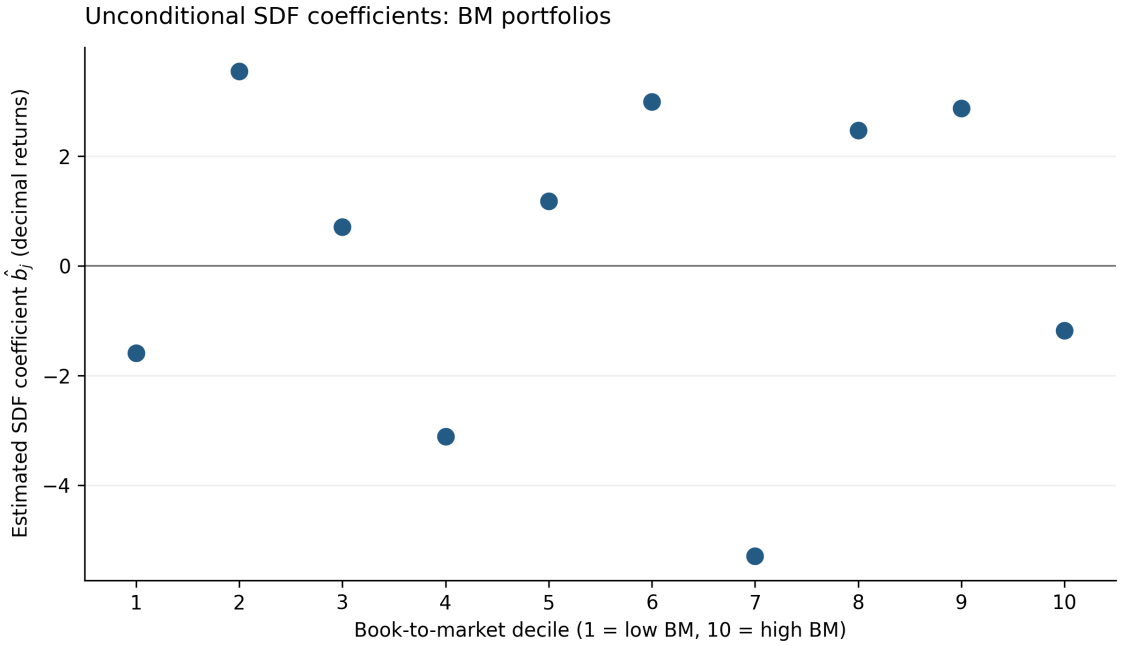



Figure 1: Estimated coefficients $\hat{b}_j$ in the unconditional SDF $\hat{M}_t^* = \hat{a} - \hat{b}' x R_t$, normalized so that $\widehat{E}[\hat{M}^*] = 1$. The sample contains 1,190 overlapping monthly observations from July 1927 through August 2026, as specified in the analysis prompt. Returns are in decimals, and the sample covariance matrix uses divisor $T$.

## (b)

Following the same approach as in part (a), I construct the unconditional linear SDF using the 10 momentum portfolios. Table 2 reports the estimated SDF coefficients.

Figure 2 shows that the estimated coefficients vary considerably across momentum deciles, with both positive and negative values. There is no clear monotonic pattern. As in part (a), this pattern reflects both the expected excess returns and the covariance structure of the portfolios.

The sample correlation between the BM and MOM SDFs is 0.3555, indicating a moderate positive relationship. This suggests that the two SDFs capture some common variation, but also reflect different components of asset returns. The sample correlation between the BM and MOM SDFs is

$$\widehat{\mathrm{Corr}}\left(\hat{M}^{*,BM}, \hat{M}^{*,MOM}\right) = 0.3555.$$

This uses 1,190 aligned monthly observations from July 1927 through August 2026.

Table 2: Estimated MOM SDF intercept and coefficients, rounded to four decimal places. Each $\hat{b}_j$ corresponds to MOM decile $j$; estimation uses decimal monthly excess returns.

| $\hat{a}$ | $\hat{b}_1$ | $\hat{b}_2$ | $\hat{b}_3$ | $\hat{b}_4$ | $\hat{b}_5$ | $\hat{b}_6$ | $\hat{b}_7$ | $\hat{b}_8$ | $\hat{b}_9$ | $\hat{b}_{10}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 1.0666 | -4.0194 | 1.1429 | 0.1024 | 1.4897 | 2.5317 | -0.8195 | 0.2777 | 0.9818 | -0.9414 | 3.4370 |

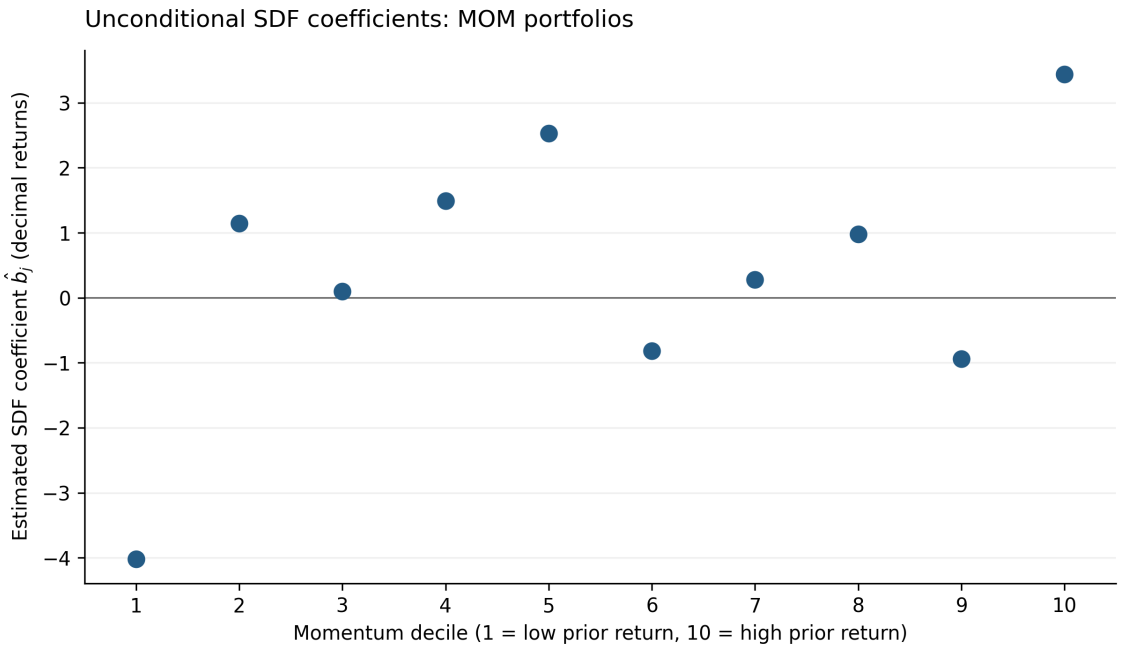



Figure 2: Estimated coefficients $\hat{b}_j$ in the unconditional SDF $\hat{M}_t^* = \hat{a} - \hat{b}' x R_t$ for the ten momentum portfolios, normalized so that $\widehat{E}[\hat{M}^*] = 1$. The sample contains 1,190 overlapping monthly observations from July 1927 through August 2026, as specified in the analysis prompt. Returns are in decimals, and the sample covariance matrix uses divisor $T$, as in part (a).

## (c)

Following the same approach as in parts (a) and (b), I construct the unconditional linear SDF using the combined 20 BM and MOM portfolios. Table 3 reports the estimated SDF coefficients.

Figure 3 shows that the estimated coefficients vary considerably across both BM and MOM deciles, with positive and negative values and no clear monotonic pattern. Compared with parts (a) and (b), the coefficients change when the two sets of portfolios are combined, reflecting the additional covariance relationships and diversification opportunities.

Table 4 reports monthly Sharpe ratios of 0.1919, 0.2581, and 0.3045 for the BM, MOM, and combined maximum Sharpe ratio portfolios, respectively. The combined portfolio achieves the highest Sharpe ratio, consistent with the theoretical prediction that expanding the investment opportunity set cannot reduce the maximum attainable Sharpe ratio.

This result is also consistent with the SDF volatility bound discussed in class. Under the normalization $E[M^*] = 1$, the volatility of the constructed SDF equals the maximum Sharpe ratio. Thus, including both BM and MOM portfolios increases the maximum attainable Sharpe ratio and the volatility of the corresponding SDF.

Table 3: Combined SDF intercept and coefficients, rounded to four decimal places. The first ten coefficients correspond to BM portfolios and the next ten to MOM portfolios. Estimation uses decimal excess returns.

| SDF intercept $\hat{a}$ | | | | | | | | | | 1.0927 |
|---|---|---|---|---|---|---|---|---|---|---|
| Decile $j$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| BM | -1.9664 | 2.4739 | 0.5781 | -3.7439 | 1.0655 | 2.6840 | -4.6456 | 2.4707 | 3.7398 | -0.2415 |
| MOM | -4.4169 | 0.6354 | -0.1712 | 1.9480 | 1.4236 | -0.8781 | -0.1639 | 0.9802 | -1.2708 | 3.4473 |

Table 4: Monthly Sharpe ratios of $xR_t^* = \hat{b}' xR_t$, using the same 1,190 months from July 1927 through August 2026. Variances use divisor $T$.

| Portfolio set | Monthly Sharpe ratio |
|---|---|
| BM | 0.1919 |
| MOM | 0.2581 |
| Combined | 0.3045 |

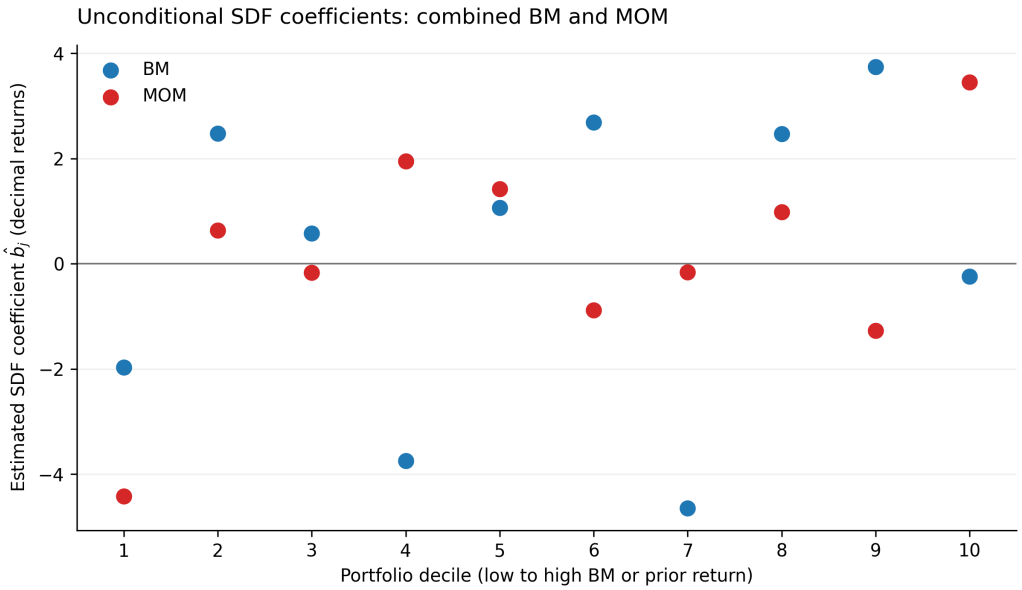



Figure 3: Coefficients in the combined BM–MOM SDF, normalized so that $\widehat{E}[\hat{M}^*] = 1$. Blue dots denote BM portfolios and red dots denote MOM portfolios. Returns are in decimals; the covariance matrix uses divisor $T$.

## (d)

Table 5 reports the pricing errors of the 20 BM and MOM portfolios under the three SDFs constructed in parts (a), (b), and (c), using

$$\hat{\alpha}_p = \hat{E}[xR_p] - \widehat{\mathrm{Cov}}(-\hat{M}^*, xR_p).$$

The pricing errors are essentially zero for the portfolios used to construct each SDF. Specifically, the BM SDF prices all 10 BM portfolios, the MOM SDF prices all 10 MOM portfolios, and the combined SDF prices all 20 portfolios, with errors close to numerical precision.

However, the BM and MOM SDFs exhibit nonzero pricing errors when applied to portfolios outside their respective construction sets. The maximum absolute pricing error is 0.010157 for MOM portfolios under the BM SDF, compared with 0.003450 for BM portfolios under the MOM SDF.

These results suggest that an SDF constructed from one set of portfolios does not necessarily price another set of portfolios accurately. The combined SDF eliminates the in-sample pricing errors by incorporating both sets of portfolios. However, this exact in-sample fit does not necessarily imply good out-of-sample pricing performance.

Table 5: Monthly pricing errors $\hat{\alpha}_p$ under the BM, MOM, and combined SDFs, in decimal return units and rounded to six decimal places. The common sample is July 1927 through August 2026 (1,190 months); covariances use divisor $T$.

| Portfolio | BM SDF | MOM SDF | Combined SDF |
|---|---|---|---|
| BM: Lo 10 | 0.000000 | -0.001511 | 0.000000 |
| BM: 2-Dec | 0.000000 | -0.000101 | 0.000000 |
| BM: 3-Dec | 0.000000 | -0.000223 | 0.000000 |
| BM: 4-Dec | 0.000000 | -0.000778 | 0.000000 |
| BM: 5-Dec | 0.000000 | 0.000504 | 0.000000 |
| BM: 6-Dec | 0.000000 | 0.001261 | 0.000000 |
| BM: 7-Dec | 0.000000 | 0.000281 | 0.000000 |
| BM: 8-Dec | 0.000000 | 0.002376 | 0.000000 |
| BM: 9-Dec | 0.000000 | 0.003450 | 0.000000 |
| BM: Hi 10 | 0.000000 | 0.003198 | 0.000000 |
| MOM: Lo PRIOR | -0.010157 | 0.000000 | 0.000000 |
| MOM: PRIOR 2 | -0.005051 | 0.000000 | 0.000000 |
| MOM: PRIOR 3 | -0.003272 | 0.000000 | 0.000000 |
| MOM: PRIOR 4 | -0.001488 | 0.000000 | 0.000000 |
| MOM: PRIOR 5 | -0.000882 | 0.000000 | 0.000000 |
| MOM: PRIOR 6 | -0.000263 | 0.000000 | 0.000000 |
| MOM: PRIOR 7 | 0.000509 | 0.000000 | 0.000000 |
| MOM: PRIOR 8 | 0.001902 | 0.000000 | 0.000000 |
| MOM: PRIOR 9 | 0.002240 | 0.000000 | 0.000000 |
| MOM: Hi PRIOR | 0.005418 | 0.000000 | 0.000000 |

## (e)

Following Question 2(c), I re-estimate the SDF using monthly excess returns of the 20 BM and MOM decile portfolios from July 1927 through December 1989:

$$\hat{M}_t^* = \hat{a} - \hat{\mathbf{b}}' \mathbf{x}\mathbf{R}_t.$$

This gives the estimated intercept $\hat{a}$ and coefficient vector $\hat{\mathbf{b}}$.

I then hold the estimated coefficients fixed and construct the out-of-sample SDF using monthly excess returns from January 1990 through August 2026:

$$\hat{M}_{OS,t}^* = \hat{a} - \hat{\mathbf{b}}' \mathbf{x}\mathbf{R}_t.$$

For each of the 20 portfolios, I calculate the out-of-sample pricing error as

$$\hat{\alpha}_{OS,p} = \widehat{E}[xR_p] - \widehat{\mathrm{Cov}}(-\hat{M}_{OS}^*, xR_p),$$

where both the expectation and covariance are estimated using only the out-of-sample observations. The resulting pricing errors are reported in Table 6. All 20 portfolios exhibit positive out-of-sample pricing errors, suggesting that the estimated SDF does not fully capture their average excess returns.

Table 6: Out-of-sample monthly pricing errors $\hat{\alpha}_{OS,p}$, in decimal return units and rounded to six decimal places. The SDF is estimated using July 1927 through December 1989 (750 months), with its intercept and coefficients held fixed for evaluation over January 1990 through August 2026 (440 months). Covariances use divisor $T_{OS}$.

| Portfolio | Out-of-sample pricing error |
|---|---|
| BM: Lo 10 | 0.010971 |
| BM: 2-Dec | 0.007858 |
| BM: 3-Dec | 0.009002 |
| BM: 4-Dec | 0.006957 |
| BM: 5-Dec | 0.007986 |
| BM: 6-Dec | 0.007178 |
| BM: 7-Dec | 0.006977 |
| BM: 8-Dec | 0.005923 |
| BM: 9-Dec | 0.008269 |
| BM: Hi 10 | 0.009709 |
| MOM: Lo PRIOR | 0.027055 |
| MOM: PRIOR 2 | 0.016483 |
| MOM: PRIOR 3 | 0.013905 |
| MOM: PRIOR 4 | 0.012077 |
| MOM: PRIOR 5 | 0.009180 |
| MOM: PRIOR 6 | 0.008297 |
| MOM: PRIOR 7 | 0.007601 |
| MOM: PRIOR 8 | 0.006444 |
| MOM: PRIOR 9 | 0.004322 |
| MOM: Hi PRIOR | 0.005453 |

# Question 3

## (a)

The investor solves

$$\max_{\varpi_t} E_t[U(W_{t+1})] \quad \text{s.t.} \quad W_{t+1} = W_t\big(R_{f,t+1} + \varpi_t' x R_{t+1}\big).$$

Assume that $W_t > 0$, $W_{t+1} > 0$ almost surely, and that the optimal portfolio choice is interior. Under the usual regularity conditions, we can derive the first-order condition by differentiating expected utility w.r.t. $\varpi_t$.

This problem can be written as

$$\max_{\varpi_t} E_t\big[U\big(W_t\big(R_{f,t+1} + \varpi_t' x R_{t+1}\big)\big)\big].$$

$$\frac{\partial E_t\big[U\big(W_t\big(R_{f,t+1} + \varpi_t' x R_{t+1}\big)\big)\big]}{\partial \varpi_t} = U'(W_{t+1}) \cdot W_t \cdot x R_{t+1}.$$

$$U(w) = \frac{w^{1-\gamma}}{1-\gamma}, \qquad U'(w) = w^{-\gamma}.$$

Thus, $U'(W_{t+1}) = W_{t+1}^{-\gamma}$.

$$\begin{aligned}\frac{\partial E_t\big[U\big(W_t\big(R_{f,t+1} + \varpi_t' x R_{t+1}\big)\big)\big]}{\partial \varpi_t} &= E_t\big[U'(W_{t+1}) \cdot W_t\big] \\ &= E_t\Big[W_{t+1}^{-\gamma} \cdot W_t \cdot x R_{t+1}\Big] = 0.\end{aligned}$$

$$E_t\left[\frac{W_t}{W_{t+1}^{\gamma}} x R_{t+1}\right] = W_t E_t\left[\frac{1}{W_{t+1}^{\gamma}} x R_{t+1}\right] = 0.$$

Since $W_{t+1}^{\gamma} = (W_t \cdot R_{w,t+1})^{\gamma}$,

$$E_t\left[\frac{1}{(W_t \cdot R_{w,t+1})^{\gamma}} x R_{t+1}\right] = W_t^{-\gamma} E_t\Big[R_{w,t+1}^{-\gamma} x R_{t+1}\Big] = 0.$$

Since $W_t > 0$, we have

$$E_t\Big[R_{w,t+1}^{-\gamma} x R_{t+1}\Big] = 0.$$

**(b)**

Table 7: GMM estimates using the supplied equity/risk-free log returns, December 1927 through December 2021. Values are rounded to four decimal places.

| Statistic | Value |
|---|---|
| Risk aversion $\hat{\gamma}$ | 1.9061 |
| Standard error $SE(\hat{\gamma})$ | 0.6026 |
| Jacobian $\hat{G}$ | -0.0480 |
| Long-run variance $\hat{S}$ | 0.9439 |
| Sample moment $\bar{g}_T(\hat{\gamma})$ | 0.0000 |
| Observations $T$ | 1129 |
| HAC lag $L$ | 26 |

## (c)

Using the identity weighting matrix, the GMM estimate of the relative risk aversion coefficient is $\hat{\gamma} = 3.0064$, with a Newey–West standard error of 0.7207. The instrument denominator is the variance of gross equity returns over the 973 estimation observations from December 1940 through December 2021.

The estimated pricing errors for the equity portfolio and the managed portfolio are $-0.0038$ and 0.0019, respectively. Neither is statistically significant.

The joint $\chi^2(1)$ test yields a p-value of 0.4276, so we fail to reject the null hypothesis that both SDF pricing conditions hold simultaneously at conventional significance levels. This suggests that the data do not provide sufficient evidence to reject the pricing restrictions implied by the power-utility SDF model.

Table 8: Identity-weighting two-moment GMM estimates, December 1940 through December 2021. Results are rounded to four decimal places.

| Statistic | Value |
|---|---|
| Risk aversion $\hat{\gamma}$ | 3.0064 |
| Standard error $SE(\hat{\gamma})$ | 0.7207 |
| Pricing error $\bar{g}_1$ | -0.0038 |
| $t$-statistic for $\bar{g}_1$ | -0.7934 |
| Pricing error $\bar{g}_2$ | 0.0019 |
| $t$-statistic for $\bar{g}_2$ | 0.7934 |
| Joint Wald $\chi^2$ statistic | 0.6294 |
| Joint-test $p$-value ($df = 1$) | 0.4276 |
| Observations $T$ | 973 |
| Common HAC lag $L$ | 24 |

## (d)

For the two-step GMM estimation, I use the inverse of the Newey–West covariance matrix estimated in Q3(c) as the weighting matrix. The resulting estimate of risk aversion is $\hat{\gamma}_2 = 2.9508$, with a standard error of 0.7106, compared with the first-step estimate of 3.0064.

The estimated pricing errors for the equity portfolio and the managed portfolio are -0.0020 and 0.0055, respectively. Neither is statistically significant. The joint $\chi^2(1)$ test yields a p-value of 0.4192, so we fail to reject the null hypothesis that both pricing errors are jointly zero at conventional significance levels.

Table 9: Two-step efficient GMM, December 1940 through December 2021. Results are rounded to four decimal places.

| Statistic | Value |
|---|---|
| Risk aversion $\hat{\gamma}_2$ | 2.9508 |
| Standard error $SE(\hat{\gamma}_2)$ | 0.7106 |
| Pricing error $\bar{g}_1(\hat{\gamma}_2)$ | -0.0020 |
| $t$-statistic for $\bar{g}_1$ | -0.8078 |
| Pricing error $\bar{g}_2(\hat{\gamma}_2)$ | 0.0055 |
| $t$-statistic for $\bar{g}_2$ | 0.8078 |
| Joint Wald $\chi^2$ statistic | 0.6525 |
| Joint-test $p$-value ($df = 1$) | 0.4192 |
| Observations $T$ | 973 |
| First-step HAC lag $L_1$ | 24 |
| Second-step HAC lag $L_2$ | 24 |
| First-step risk aversion $\hat{\gamma}_1$ | 3.0064 |

## (e)

Table 10 reports the identity-weighted GMM estimation results using the equity portfolio and 20 book-to-market (BM) and momentum (MOM) decile portfolios.

To ensure consistency with the annual real returns in the EQ Dataset, I compound the monthly nominal returns of the BM and MOM portfolios and the risk-free rate over 12 months, calculate the annual nominal excess returns, and then deflate them using the corresponding 12-month CPI inflation factor. The sample covers December 1927 through December 2021.

Several individual pricing errors are statistically significant, particularly among the momentum portfolios, suggesting that the power-utility SDF does not price all portfolios equally well.

The joint $\chi^2(20)$ test yields a p-value of 0.0000, so we reject the null hypothesis that all 21 pricing errors are jointly zero at conventional significance levels.

Table 10: Identity-weighting GMM with 21 moments. Annual real returns observed monthly, December 1927 through December 2021. Results are rounded to four decimal places. BM and MOM numbers denote deciles from low to high.

| Statistic | | Value |
|---|---|---|
| Risk aversion $\hat{\gamma}$ | | 1.9314 |
| Standard error $SE(\hat{\gamma})$ | | 0.6093 |
| Joint Wald $\chi^2(20)$ statistic | | 202.9263 |
| Joint-test $p$-value | | 0.0000 |
| Observations $T$ | | 1129 |
| Common HAC lag $L$ | | 26 |
| **Portfolio** | **Pricing error** | ***t*-statistic** |
| Equity | -0.0012 | -0.3150 |
| BM1 | -0.0079 | -0.7976 |
| BM2 | 0.0108 | 1.5213 |
| BM3 | 0.0031 | 0.5205 |
| BM4 | -0.0022 | -0.3897 |
| BM5 | 0.0076 | 1.0870 |
| BM6 | 0.0135 | 1.6334 |
| BM7 | -0.0021 | -0.2321 |
| BM8 | 0.0208 | 2.6781 |
| BM9 | 0.0317 | 3.4069 |
| BM10 | 0.0175 | 1.2359 |
| MOM1 | -0.0984 | -7.9325 |
| MOM2 | -0.0426 | -5.1481 |
| MOM3 | -0.0299 | -3.1821 |
| MOM4 | -0.0119 | -1.6900 |
| MOM5 | -0.0009 | -0.1448 |
| MOM6 | -0.0004 | -0.0863 |
| MOM7 | 0.0109 | 1.5880 |
| MOM8 | 0.0242 | 4.3724 |
| MOM9 | 0.0295 | 3.7306 |
| MOM10 | 0.0611 | 4.7192 |

**Additional check: common estimation sample**

I noticed that the estimated risk-aversion coefficient $\hat{\gamma}$ varies considerably across Q3(b)–Q3(e). In Q3(b) and Q3(e), the estimates are 1.9061 and 1.9314, respectively, while Q3(c) and Q3(d) produce higher estimates of 3.0064 and 2.9508. One possible explanation is the difference in estimation periods: Q3(b) and Q3(e) use December 1927 through December 2021, whereas Q3(c) and Q3(d) use December 1940 through December 2021.

To investigate this, I re-estimate Q3(b) and Q3(e) using the restricted sample from December 1940 through December 2021, keeping their original moment conditions, weighting matrices, and estimation methods unchanged. Table 11 reports the comparison.

After restricting the sample, the estimated $\hat{\gamma}$ increases from 1.9061 to 2.8878 for Q3(b), and from 1.9314 to 3.0359 for Q3(e). The estimates across all four specifications are now much closer, ranging from 2.8878 to 3.0359. This suggests that differences in the estimation periods account for a substantial part of the variation in $\hat{\gamma}$, while the remaining differences may reflect the different moment conditions and GMM weighting matrices.

Table 11: Additional sample comparison for Question 3. Original samples run from December 1927 through December 2021; restricted samples and existing Q3(c)/(d) results use December 1940 through December 2021. Estimates and standard errors are rounded to four decimal places.

| Model | Sample | Moments | Weight | $T$ | $L$ | $\hat{\gamma}$ | $SE$ |
|---|---|---|---|---|---|---|---|
| Q3(b) | Original | 1 | $I$ | 1129 | 26 | 1.9061 | 0.6026 |
| Q3(b) | Restricted | 1 | $I$ | 973 | 24 | 2.8878 | 0.7086 |
| Q3(c) | Existing | 2 | $I$ | 973 | 24 | 3.0064 | 0.7207 |
| Q3(d) | Existing | 2 | $\hat{S}_1^{-1}$ | 973 | 24 | 2.9508 | 0.7106 |
| Q3(e) | Original | 21 | $I$ | 1129 | 26 | 1.9314 | 0.6093 |
| Q3(e) | Restricted | 21 | $I$ | 973 | 24 | 3.0359 | 0.7088 |

Only Q3(b) and Q3(e) are re-estimated on the restricted sample. Their original moment conditions, return construction, identity weights, optimization methods, and HAC rules are retained. Annual returns are constructed before restricting dates; HAC lags are selected anew on the restricted moments. Q3(c) and Q3(d) are copied from their saved results; the Q3(d) lag shown is the second-step lag. $I$ denotes the identity matrix of the appropriate dimension. All original results are retained.

# Question 4

## (a)

By applying a first-order Taylor approximation, we have

$$\begin{aligned} M_{t+1} &= f(F_{t+1}, \theta) \\ &\approx f(E_t[F], \theta) + \partial_F f(E_t[F], \theta)'\big(F_{t+1} - E_t[F]\big) \equiv M_{t+1}^{\text{linear}}. \end{aligned}$$

Since $M_{t+1} = R_{w,t+1}^{-\gamma}$,

$$F_{t+1} = R_{w,t+1}, \qquad \theta = \gamma, \qquad f(F, \gamma) = F^{-\gamma}.$$

Substituting the actual form, we have

$$\begin{aligned} f\big(E_t[R_{w,t+1}], \gamma\big) &= \big(E_t[R_{w,t+1}]\big)^{-\gamma}, \\ f'\big(E_t[R_{w,t+1}], \gamma\big) &= -\gamma\big(E_t[R_{w,t+1}]\big)^{-\gamma-1}. \end{aligned}$$

$$\begin{aligned} M_{t+1} = R_{w,t+1}^{-\gamma} &\approx \big(E_t[R_{w,t+1}]\big)^{-\gamma} - \gamma\big(E_t[R_{w,t+1}]\big)^{-\gamma-1}\big(R_{w,t+1} - E_t[R_{w,t+1}]\big) \\ &= \big(E_t[R_{w,t+1}]\big)^{-\gamma} - \gamma\big(E_t[R_{w,t+1}]\big)^{-\gamma-1} R_{w,t+1} + \gamma\big(E_t[R_{w,t+1}]\big)^{-\gamma} \\ &= (1+\gamma)\big(E_t[R_{w,t+1}]\big)^{-\gamma} - \gamma\big(E_t[R_{w,t+1}]\big)^{-\gamma-1} R_{w,t+1}. \end{aligned}$$

$$a_t = (1+\gamma)\big(E_t[R_{w,t+1}]\big)^{-\gamma}, \qquad b_t = \gamma\big(E_t[R_{w,t+1}]\big)^{-\gamma-1}.$$

Thus,

$$M_{t+1} \approx a_t - b_t R_{w,t+1}, \qquad a_t = (1+\gamma)\big(E_t[R_{w,t+1}]\big)^{-\gamma}, \qquad b_t = \gamma\big(E_t[R_{w,t+1}]\big)^{-\gamma-1}.$$

For $\gamma > 0$, $a_t - b_t R_{w,t+1} < 0$ if

$$\begin{aligned} (1+\gamma)\big(E_t[R_{w,t+1}]\big)^{-\gamma} - \gamma\big(E_t[R_{w,t+1}]\big)^{-\gamma-1} R_{w,t+1} &< 0, \\ (1+\gamma)\big(E_t[R_{w,t+1}]\big)^{-\gamma} &< \gamma\big(E_t[R_{w,t+1}]\big)^{-\gamma-1} R_{w,t+1}, \\ R_{w,t+1} &> \frac{(1+\gamma)\big(E_t[R_{w,t+1}]\big)^{-\gamma}}{\gamma\big(E_t[R_{w,t+1}]\big)^{-\gamma-1}} = \frac{1+\gamma}{\gamma} E_t[R_{w,t+1}]. \end{aligned}$$

Thus, the linear SDF is not guaranteed to remain positive, unlike the original nonlinear SDF. This is a disadvantage because strict positivity of an SDF is sufficient to ensure no-arbitrage pricing. The linear approximation does not necessarily preserve this property.

## (b)

Table 12 reports the identity-weighted GMM estimation results for the normalized linear SDF using the equity portfolio and 20 book-to-market (BM) and momentum (MOM) decile portfolios.

Following the same construction of annual real excess returns as in Question 3(e), the estimated SDF coefficient is $\hat{b} = 1.9356$, with a standard error of 0.6425.

Several individual pricing errors are statistically significant, particularly among the momentum portfolios, suggesting that the linear SDF does not price all portfolios equally well.

The joint $\chi^2(20)$ test yields a statistic of 231.3234 and a p-value of 0.0000, so we reject the null hypothesis that all 21 pricing errors are jointly zero at conventional significance levels.

Table 12: Identity-weighting GMM for the normalized linear SDF, December 1927 through December 2021. Results are rounded to four decimal places; standard errors account for estimation of the mean wealth return. BM and MOM numbers denote deciles from low to high.

| Statistic | | Value |
|---|---|---|
| SDF slope $\hat{b}$ | | 1.9356 |
| Standard error $SE(\hat{b})$ | | 0.6425 |
| Joint Wald $\chi^2(20)$ statistic | | 231.3234 |
| Joint-test $p$-value | | 0.0000 |
| Observations $T$ | | 1129 |
| Common HAC lag $L$ | | 26 |
| **Portfolio** | **Pricing error** | **$t$-statistic** |
| Equity | 0.0025 | 0.5249 |
| BM1 | -0.0041 | -0.3663 |
| BM2 | 0.0168 | 2.0332 |
| BM3 | 0.0064 | 0.9379 |
| BM4 | -0.0023 | -0.3875 |
| BM5 | 0.0101 | 1.4719 |
| BM6 | 0.0149 | 1.9235 |
| BM7 | -0.0027 | -0.2914 |
| BM8 | 0.0178 | 2.1771 |
| BM9 | 0.0252 | 2.5033 |
| BM10 | 0.0069 | 0.4700 |
| MOM1 | -0.1076 | -8.8135 |
| MOM2 | -0.0484 | -5.5359 |
| MOM3 | -0.0275 | -2.8888 |
| MOM4 | -0.0131 | -1.8467 |
| MOM5 | 0.0005 | 0.0673 |
| MOM6 | 0.0002 | 0.0337 |
| MOM7 | 0.0163 | 2.2130 |
| MOM8 | 0.0291 | 4.5403 |
| MOM9 | 0.0356 | 3.9909 |
| MOM10 | 0.0704 | 5.0555 |

## (c)

Table 13 reports the two-step efficient GMM estimation results for the normalized linear SDF using the equity portfolio and 20 book-to-market (BM) and momentum (MOM) decile portfolios.

Using the inverse of the estimated Newey–West long-run covariance matrix as the weighting matrix, the estimated SDF coefficient is $\hat{b} = 1.6123$, with a standard error of 0.3925. Compared with the identity-weighted GMM results in Question 4(b), both the coefficient estimate and its standard error decrease.

Several individual pricing errors remain statistically significant, particularly among the momentum portfolios, suggesting that the linear SDF does not price all portfolios equally well.

The efficient-GMM $J$-test yields a $\chi^2(20)$ statistic of 231.3234 and a p-value of 0.0000, so we reject the null hypothesis that all 21 pricing errors are jointly zero at conventional significance levels. This conclusion is consistent with the identity-weighted GMM results in Question 4(b).

Table 13: Two-step efficient GMM for the normalized linear SDF, December 1927 through December 2021. Results are rounded to four decimal places; inference accounts for estimation of the mean wealth return. Both joint tests use 20 degrees of freedom.

| Statistic | | Value |
|---|---|---|
| SDF slope $\hat{b}_{4c}$ | | 1.6123 |
| Standard error $SE(\hat{b}_{4c})$ | | 0.3925 |
| Efficient-GMM $J$ statistic | | 231.3234 |
| $J$-test $p$-value | | 0.0000 |
| Observations $T$ | | 1129 |
| Second-step HAC lag $L_2$ | | 26 |
| Covariance-adjusted Wald $\chi^2(20)$ | | 241.4446 |
| Adjusted Wald $p$-value | | 0.0000 |
| **Portfolio** | **Pricing error** | **$t$-statistic** |
| Equity | 0.0161 | 0.7936 |
| BM1 | 0.0101 | 0.4485 |
| BM2 | 0.0294 | 1.5998 |
| BM3 | 0.0195 | 1.0873 |
| BM4 | 0.0126 | 0.5426 |
| BM5 | 0.0242 | 1.0083 |
| BM6 | 0.0297 | 1.1616 |
| BM7 | 0.0128 | 0.4805 |
| BM8 | 0.0348 | 1.3980 |
| BM9 | 0.0434 | 1.7947 |
| BM10 | 0.0280 | 0.9487 |
| MOM1 | -0.0875 | -2.9355 |
| MOM2 | -0.0306 | -1.1193 |
| MOM3 | -0.0126 | -0.4535 |
| MOM4 | 0.0023 | 0.0905 |
| MOM5 | 0.0142 | 0.6664 |
| MOM6 | 0.0147 | 0.6565 |
| MOM7 | 0.0293 | 1.3663 |
| MOM8 | 0.0425 | 2.1500 |
| MOM9 | 0.0496 | 2.4055 |
| MOM10 | 0.0854 | 3.9235 |

(d)

Table 14: Time-series regressions on gross $R_w$, December 1927 through December 2021. Values are rounded to four decimal places. Individual intercept tests use HAC; the GRS test uses its conventional $F$ reference.

| Statistic | Value |
|---|---|
| Requested $\hat{\lambda} = \bar{R}_w$ | 1.0877 |
| Newey–West $SE(\hat{\lambda})$ | 0.0178 |
| Conventional GRS $F$ statistic | 1589.2178 |
| GRS degrees of freedom | (21, 1107) |
| GRS $p$-value | 0.0000 |
| Observations $T$ | 1129 |
| HAC lag for $\hat{\lambda}$ | 26 |

| Portfolio | $\hat{\alpha}$ | $SE(\hat{\alpha})$ | $t_\alpha$ | $\hat{\beta}$ | $L_j$ |
|---|---|---|---|---|---|
| Equity | -0.9994 | 0.0327 | -30.5350 | 0.9960 | 28 |
| BM1 | -1.0492 | 0.0435 | -24.1289 | 1.0390 | 27 |
| BM2 | -0.9101 | 0.0571 | -15.9392 | 0.9215 | 26 |
| BM3 | -0.9594 | 0.0337 | -28.4702 | 0.9601 | 27 |
| BM4 | -1.1027 | 0.1157 | -9.5297 | 1.0939 | 26 |
| BM5 | -1.0235 | 0.0870 | -11.7626 | 1.0275 | 27 |
| BM6 | -1.0782 | 0.1193 | -9.0359 | 1.0866 | 26 |
| BM7 | -1.1439 | 0.1427 | -8.0185 | 1.1345 | 26 |
| BM8 | -1.2376 | 0.1444 | -8.5710 | 1.2480 | 26 |
| BM9 | -1.3150 | 0.1683 | -7.8151 | 1.3323 | 25 |
| BM10 | -1.5519 | 0.2222 | -6.9844 | 1.5496 | 26 |
| MOM1 | -1.5837 | 0.1218 | -13.0003 | 1.4675 | 25 |
| MOM2 | -1.3570 | 0.1446 | -9.3872 | 1.3009 | 23 |
| MOM3 | -1.1210 | 0.0856 | -13.0936 | 1.0871 | 26 |
| MOM4 | -1.1487 | 0.1345 | -8.5371 | 1.1289 | 25 |
| MOM5 | -1.0102 | 0.0844 | -11.9628 | 1.0047 | 25 |
| MOM6 | -1.0664 | 0.1006 | -10.6019 | 1.0603 | 26 |
| MOM7 | -0.9474 | 0.0492 | -19.2655 | 0.9580 | 27 |
| MOM8 | -0.9611 | 0.0339 | -28.3310 | 0.9843 | 25 |
| MOM9 | -0.9968 | 0.0408 | -24.4483 | 1.0263 | 26 |
| MOM10 | -1.0282 | 0.1189 | -8.6497 | 1.0922 | 25 |

The sample contains 1,129 monthly observations of annual real returns from December 1927 through December 2021, identical to Q4(b). The requested constant-only regression uses gross $R_w$ and reports its sample mean as $\hat{\lambda}$. The lambda-series pilot is $m = 7$, $C = 2.4496$, and $L = \lceil CT^{1/3} \rceil = \lceil 25.5067 \rceil = 26$. Each portfolio regression selects its own common lag from the sum of the two centered score components. All HAC calculations retain Bartlett weights, $T - \ell$ divisors, and ceiling pilot/final rounding.

The regression intercepts and the normalized-SDF pricing errors have different definitions:

$$\hat{\alpha}_j^{OLS} = \hat{\mu}_j - \hat{\beta}_j \bar{R}_w, \qquad \hat{e}_j^{SDF} = \hat{\mu}_j - \hat{\beta}_j \hat{\lambda}_{SDF}, \qquad \hat{\lambda}_{SDF} = \hat{b}_{4b} \widehat{\mathrm{Var}}(R_w).$$

Here $\bar{R}_w = 1.087734$, whereas $\hat{\lambda}_{SDF} = 0.081817$. The reported alphas are the requested gross-factor OLS intercepts, not the Q4(b) SDF errors.

At the 5% level, 21 of the 21 intercepts have two-sided asymptotic-normal HAC $p$-values below 0.05. The conventional GRS reference yields $p \approx 1.986 \times 10^{-807}$ and rejects the joint zero-intercept null at that level. The residual covariance has full rank 21 and condition number 750.09; all 21 assets are retained, and its ordinary inverse is used with divisor $T$. The GRS $F$ calibration relies on the conventional GRS assumptions; it is separate from the HAC corrections used for individual intercepts. Annual returns observed in overlapping monthly windows have serial dependence.

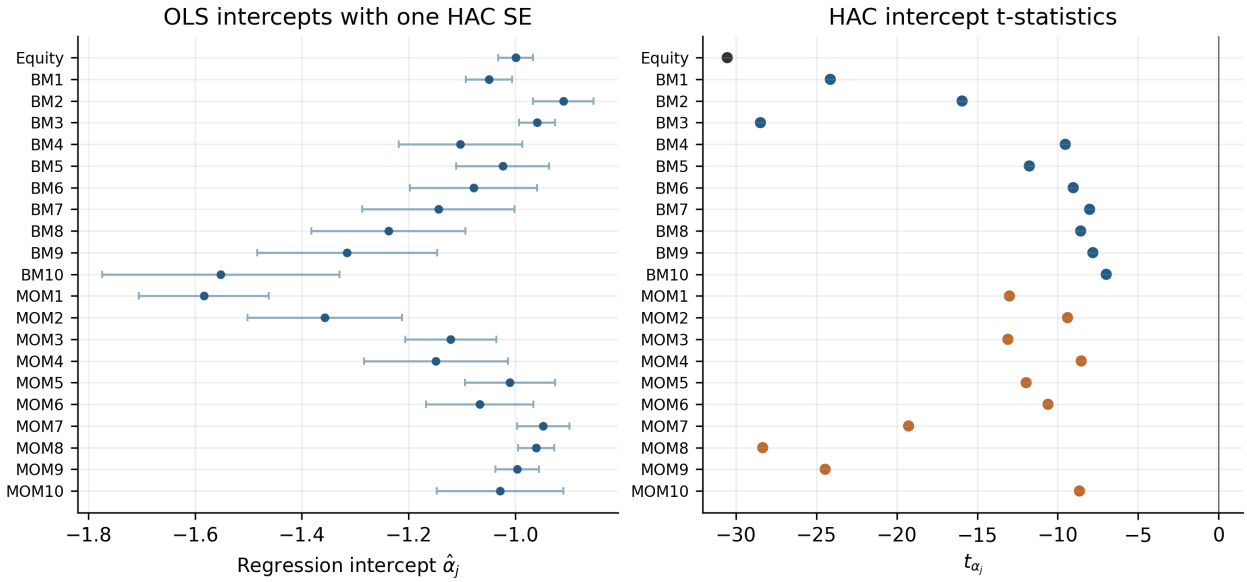



Figure 4: Intercept estimates and HAC $t$-statistics from the requested regressions of annual real excess returns on a constant and gross $R_w$. Error bars show one HAC standard error.